4 **(a)** A small ball rests at point P on a curved track of radius r, as shown in Figure 4.1.

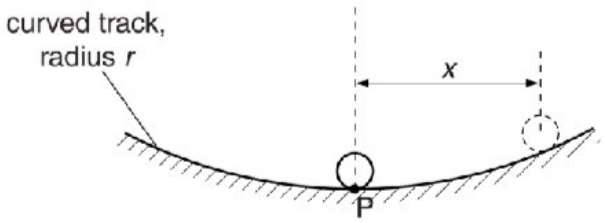


Figure 4.1

The ball is moved a small distance to one side and is then released. The horizontal displacement x of the ball is related to its acceleration a towards P by the expression

$$a = -\frac{gx}{r}$$

where g is the acceleration of free fall.

(i) Show that the ball undergoes simple harmonic motion.

.....

.....

.....

..... [2]

(ii) The radius r of curvature of the track is 28 cm.

Determine the time interval τ between the ball passing point P and then returning to point P.

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **4(a)** is shown in Figure 4.2.

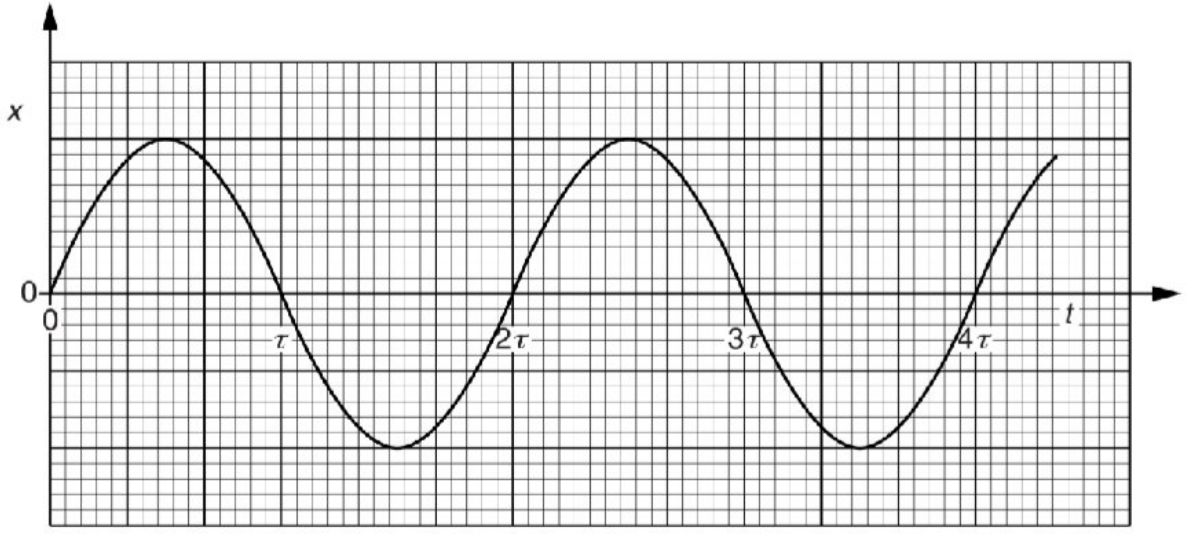


Figure 4.2

Some moisture now forms on the track, causing the ball to come to rest after approximately 15 oscillations.

On the axes of Figure 4.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. Assume the moisture forms at time $t = 0$.

[3]

[Total: 8]